

USENGIMANA RICHARD

Data Scientist | Data Analyst | Machine Learning Engineer

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ABOUT ME

Data Scientist and Mathematical Sciences graduate specializing in Data Science, with a strong foundation in applied mathematics, statistical modelling, machine learning, and spatial analysis.

Currently working as a Data Scientist Intern at I&M Bank Rwanda, contributing to customer segmentation and next best action modelling using unsupervised machine learning.

Experienced in transforming complex datasets into meaningful insights through data cleaning, statistical analysis, visualization, and predictive modelling. Passionate about applying data science to real-world problems in banking, public health, business, and development.

Committed to producing analytical work that is reproducible, well-documented, and decision-relevant.

PROFESSIONAL EXPERIENCE

Data Scientist Intern

I&M Bank Rwanda

📅 July 2026 – Present 📍 Kigali, Rwanda

- Working on a customer analytics project titled **Customer Segmentation and Next Best Action Modeling**.
- Applying unsupervised machine learning techniques to segment the bank's customer base into meaningful groups based on customer characteristics and behaviour.
- Supporting the identification of distinct customer segments to improve customer understanding and data-driven decision-making.
- Contributing to analytical workflows involving data preparation, exploratory analysis, modelling, and interpretation of customer segments.

Internship in Planning, Monitoring & Evaluation

Muhanga District

📅 2024 – 1 month 📍 Muhanga, Rwanda

- Gained practical experience in planning, monitoring, and evaluation activities.
- Supported data management and reporting processes related to population and administrative systems.
- Developed an understanding of government reporting, local planning, and public service delivery.

Django Web Development Training

Solvit Africa

📅 2025 – 3 months 📍 Sonatube-Remera, Rwanda

- Completed intensive training in Django web development and Python-based applications.
- Built dynamic and data-driven web applications with database integration.
- Gained practical experience in APIs, user authentication, front-end integration, and scalable web solutions.

EDUCATION

Master of Science in Mathematical Sciences
Specialization in Data Science

African Institute for Mathematical Sciences (AIMS Rwanda)

📅 2025 – 2026 📍 Kigali-Rwanda

- Specialized in Data Science, Machine Learning, Statistical Analysis, and Artificial Intelligence.
- Completed research-focused training with practical applications to real-world data problems.
- Recipient of the Mastercard Scholarship.

Bachelor of Science in Applied Mathematics

University of Rwanda

📅 2022 – 2025 📍 Kigali, Rwanda

- Focused on mathematical modelling, statistics, and computational methods.
- Recipient of the Government of Rwanda Scholarship.
- Final project focused on differential equations and mathematical analysis.

TECHNICAL SKILLS

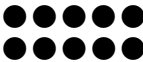
Python R SQL PostgreSQL
Machine Learning Statistical Analysis
Data Cleaning Data Visualization
Spatial Analysis Regression Models
Research Writing Microsoft Excel
Power BI GitHub Linux LaTeX

TOOLS

Pandas NumPy Scikit-learn
GeoPandas RStudio ggplot2 sf
spdep Django R Shiny Power BI
Excel

LANGUAGES

Kinyarwanda
English



RESEARCH EXPERIENCE

Spatial Analysis of Child Malnutrition in Rwanda

Master's Thesis – AIMS Rwanda

📅 2025 – 2026 📍 Rwanda

- Conducted district-level spatial analysis of child malnutrition using Rwanda Demographic and Health Survey data.
- Integrated environmental and socio-demographic factors, including rainfall, NDVI, altitude, maternal education, household wealth, water source, and sanitation.
- Applied descriptive analysis, spatial autocorrelation, regression models, and spatial regression techniques.
- Used R and Python for data cleaning, visualization, spatial analysis, mapping, and statistical modelling.
- Investigated spatial patterns, hotspots, and potential factors associated with stunting, wasting, and underweight among children under five across Rwanda's 30 districts.

SELECTED PROJECTS

Relational Database System for Insurance Policy and Claims Management

AIMS Rwanda – Database and Data Management

- Designed and implemented a normalized relational database system for insurance customers, policies, premiums, and claims processing.
- Modelled ten core entities, including Policyholder, Policy, Claim, Adjuster, Payout, and Audit Log.
- Developed an Entity-Relationship diagram and transformed it into a relational schema.
- Implemented the database using SQL, including table creation, foreign-key constraints, and sample queries.

AI Nutrition Guidance Platform

SOLVIT Hackathon

- Worked collaboratively on an AI-based platform providing early nutrition guidance for Rwandans.
- Applied data-driven and artificial intelligence approaches to support community health and nutrition awareness.
- Project selected among the top 15 teams to continue in the competition.

Intelligent Data Cleaning System

Personal Project

- Developed an AI-powered system for automated data cleaning and preprocessing.
- Created tools for data quality assessment, missing-value handling, outlier detection, and inconsistent data-format identification.
- Designed a user-friendly interface to help users prepare datasets for analysis and machine learning.

RESEARCH INTERESTS

- Machine Learning and Artificial Intelligence
- Customer Analytics and Predictive Modelling
- Statistical and Spatial Data Analysis
- Data Science for Public Health
- Business Intelligence and Decision Analytics
- Applied Mathematics and Computational Modelling

REFEREES

Dr. Lema Seknewna

@ Data Scientist & Professor

✉ AIMS Rwanda

lema.seknewna@aims.ac.rw

Current supervisor for my Master's thesis. Can attest to my skills in data analysis and Python.

Dr. Stephen Nyaranga

@ Researcher

✉ AIMS RIC

snyaranga@aimsric.org

My Co-supervisor for my Master's thesis. Can attest to my skills in Research.

Nyagahakwa Venuste

@ Lecturer

✉ University of Rwanda

venustinyo23@gmail.com

Academic reference from undergraduate studies. Can attest to my mathematical and analytical abilities.